MAHATHMA GANDHI UNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS) Regular Examination October 2025

THIRD SEMESTER Value Addition Course (VAC)-MG3VACFSQ202-DISASTER MANAGEMENT

(2024 ADMISSION ONWARDS)

ANSWER SCHEME

PART A

Multiple Choice Questions

Answer **All** Question

Each Question carries **1** Mark

1. C Spreads across countries or continents
2. B Disaster management authorities and public
3. C Farmers, treaders, teachers, children, women and elderly
4. D Flood
5. A Panic and misinformation spreading quickly
6. B Social support system and counselling
7. B Vulnerability
8. B Incident command system
9. B Lessons learned strengthen future disaster plan
10. C Bio terrorism
11. D Secrecy
12. C Begin early and be patient centered
13. A Man -made
14. B Safety and security measures
15. D Movement of tectonic plate
16. A Giving trauma counselling to survivors
17. D Disaster preparedness and risk education
18. B Track progress and apply lessons for future resilience
19. C Chief Minister of State
20. D All of the above

(20x1 mark=20)

PART B

Short Essay Questions

*Answer Any **5** Questions*

Each Question carries **4** Marks

21. Definition (1 mark)

A **landslide** is the downward movement of rock, soil, and debris along a slope under the influence of gravity. It is a common natural hazard in hilly or mountainous regions and can be triggered by natural or human factors.

Causes of Landslides(1 mark)

Natural Causes

Heavy Rainfall, Earthquakes, Volcanic Activity, Weathering: Snowmelt or Rapid Water Flow, SteepSlopes.

Human-Induced Causes

Deforestation, UnplannedConstruction, Mining and Quarrying, Poor Drainage Systems, Overgrazing

Effects of Landslides (2 marks)

. Environmental Effects

- Destruction of forests, vegetation, and wildlife habitats.
- Siltation of rivers and lakes, altering ecosystems.
- Increased soil erosion and loss of fertile topsoil.

Economic Effects

- Damage to infrastructure such as roads, bridges, railways, and buildings.
- Loss of agricultural land and decreased productivity.
- High costs of rehabilitation and reconstruction.

C. Social Effects

- Loss of human lives and livestock.
- Displacement of communities and homelessness.
- Disruption of communication and transportation networks.

Management and Mitigation of Landslides

Preventive Measures

1. **Afforestation:**
   Planting trees to stabilize soil through root binding.
2. **Proper Drainage:**
   Constructing surface and subsurface drains to prevent water accumulation.
3. **Slope Engineering:**
   Reducing slope angle, terracing, or constructing retaining walls.
4. **Soil Nailing and Rock Bolting:**
   Strengthening weak slopes with steel rods or anchors.

5. **Land Use Planning:**
   Restricting construction in landslide-prone zones.
6. **Retaining Structures:**
   Building gabion walls, check dams, or crib walls to support slopes.

Monitoring and Early Warning

- Use of **remote sensing**, **GIS**, and **rainfall thresholds** to predict potential slides.
- Installation of ground sensors to detect slope movement.
- Establishment of early warning systems for vulnerable communities.

Post-Landslide Management

- Quick **rescue and relief** operations.
- **Rehabilitation and relocation** of affected people.
- Restoration of roads, communication, and essential services.
- Long-term **slope stabilization projects**.

22. Role of Media (1 marks)
    The media plays a crucial role in reducing panic and ensuring orderly evacuation during disasters. By providing accurate, timely, and clear information, it helps people understand the situation and follow safety instructions. Media channels such as television, radio, social media, and newspapers guide the public on safe routes, shelter locations, and emergency contacts.

    Role of Media in Reducing Panic During Evacuation (3 marks)

    Responsible reporting by media prevents the spread of rumors and misinformation, which can create fear and confusion. By broadcasting messages from authorities and experts, media builds trust and confidence among people. In summary, effective communication through media ensures calm behavior, coordinated evacuation, and protection of lives during emergencies.

23. **Definition (1 mark)**
    Rebuilding refers to the process of reconstructing and restoring damaged infrastructure, housing, and essential services after a disaster. It includes repairing roads, schools, hospitals, and communication systems to restore normal functioning in the community.

    Resilience means the capacity of individuals, communities, and systems to resist, absorb, adapt to, and recover from the effects of disasters quickly and effectively. Building resilience ensures that communities are better prepared to face future hazards and can recover faster with less damage.

    Difference (3 marks)

Rebuilding

1. **Infrastructure Restoration:** Repairing damaged roads, bridges, hospitals, and schools.
2. **Housing Reconstruction:** Providing safe and disaster-resistant housing for affected families.
3. **Livelihood Recovery:** Supporting employment opportunities, agriculture, and small businesses.
4. **Social and Psychological Rehabilitation:** Offering mental health support, education, and community rebuilding programs.
5. **Environmental Restoration:** Replanting trees, restoring ecosystems, and managing natural resources.

Resilience

1. **Community Preparedness:** Training people in disaster response, first aid, and evacuation methods.
2. **Early Warning Systems:** Using technology for timely alerts and risk communication.
3. **Disaster-Resistant Infrastructure:** Constructing buildings and facilities that can withstand future hazards.
4. **Education and Awareness:** Promoting disaster education in schools and communities.
5. **Policy and Planning:** Developing effective disaster management policies that emphasize prevention and mitigation.
6. **Participation of Local Communities:** Empowering people to take active roles in planning and decision-making.

24. Definition (1 mark)

Hydrometeorological disasters are natural hazards caused by water and atmospheric processes. They include events such as floods, cyclones, droughts, storms, and heat waves. These disasters result from extreme weather conditions like heavy rainfall, strong winds, temperature variations, or prolonged dry spells.

Summary of Hydrometeorological Disasters(3 marks)

They can lead to loss of lives, destruction of property, agricultural damage, and disruption of essential services. Effective management involves accurate weather forecasting, early warning systems, disaster preparedness, and community awareness. In short, hydrometeorological disasters are weather-related events that require timely monitoring and preventive measures to reduce their impact on people and the environment.

25. Basic Structure of the Incident Command System (ICS)
Definition (1 mark)

The Incident Command System (ICS) is a standardized organizational structure used for managing emergencies efficiently. It ensures clear communication, coordination, and command during disaster response.

The basic structure (3 marks)
Incident Commander (IC):
The overall leader responsible for managing the entire operation.
Command Staff:
Public Information Officer – handles media and public communication.

Safety Officer – ensures safety of responders.

Liaison Officer – coordinates with other agencies.

General Staff:

Operations Section: Executes field response actions.

Planning Section: Collects data, prepares action plans.

Logistics Section: Provides resources and support.

Finance/Administration Section: Manages funds, records, and costs.

26. Legal Framework of Disaster Management in India

Definition (1 mark)

The Disaster Management Act, 2005 forms the main legal framework for disaster management in India. It provides a structured approach for prevention, preparedness, response, and recovery during disasters.

Key components (3 marks)

- NDMA (National Disaster Management Authority): Headed by the Prime Minister, formulates national policies and plans.
- SDMA (State Disaster Management Authority): Headed by the Chief Minister, manages disasters at the state level.
- DDMA (District Disaster Management Authority): Headed by the District Collector, handles local-level disaster management.
- NDRF (National Disaster Response Force): Special force for rescue and relief operations.
- NIDM (National Institute of Disaster Management): Provides training and research.

27. Effects of Disasters on Elderly People
Definition Disaster(1 mark)
Explanation (3 marks)

Elderly people are among the most vulnerable during disasters due to physical, emotional, and social limitations. They may face:

- Health problems such as injury, illness, or worsening of chronic diseases.
- Mobility issues, making evacuation and access to relief difficult.
- Emotional stress, anxiety, and trauma due to loss of family or property.
- Social isolation and neglect during relief and recovery phases.
- Lack of access to medicine, nutrition, and medical care in emergency shelters.

(5 x 4marks=20)

PART C

Long Essay Questions

Answer Any **1** Question

Each Question carries **10** Marks

28. Introduction (1 mark)

- A **disaster** is a sudden event that causes **serious disruption** to society, resulting in **loss of life, property, and environment**.It can be **natural** (like floods, earthquakes, or cyclones) or **man-made** (like industrial accidents or chemical spills).Disasters affect communities, economies, and the environment, demanding coordinated management and recovery efforts.

Definition (any definition)(1 mark)

"A disaster is a serious disruption of the functioning of a community or society involving widespread human, material, economic or environmental losses and impacts."

Types of Disasters(2 marks)

A. Natural Disasters

- **Hydrometeorological:** Floods, droughts, cyclones, storms.
- **Geological:** Earthquakes, volcanic eruptions, landslides.
- **Biological:** Epidemics, pandemics, pest attacks.

B. Man-Made Disasters

- Industrial accidents, fires, nuclear leaks, terrorism, deforestation, and pollution.

Concepts Related to Disaster(3 mark)

1. **Hazard:**
   A potential threat that can cause harm (e.g., earthquake, flood).
2. **Vulnerability:**
   The degree to which people or property are likely to be affected by hazards.

3. **Risk:**
   The probability of harmful consequences resulting from interactions between hazards and vulnerable conditions.
4. **Capacity:**
   The strengths and resources available to cope with or reduce disaster impact.
5. **Disaster Risk Reduction (DRR):**
   Measures and actions taken to minimize disaster risks through prevention, mitigation, and preparedness.

Effects of Disasters(2 mark)

- **Human Impact:** Loss of life, injuries, displacement.
- **Economic Impact:** Damage to property, infrastructure, and livelihoods.
- **Environmental Impact:** Soil erosion, deforestation, pollution.
- **Social Impact:** Psychological trauma, loss of community ties.

Phases of Disaster Management (1 mark)

1. **Mitigation:** Reducing potential impacts (e.g., building codes, awareness).
2. **Preparedness:** Planning and training before a disaster.
3. **Response:** Immediate action during and after disaster (rescue, relief).
4. **Recovery:** Rehabilitation and reconstruction for long-term restoration

29. Introduction(2 marks)

Media plays a vital role in **disaster planning and management** by spreading information, educating the public, and promoting preparedness. It includes **television, radio, newspapers, internet, and social media**, which help reach people quickly and effectively before, during, and after disasters.

Advantages of Using Media (4 marks)

Quick Dissemination of Information:
Media spreads early warnings and safety instructions rapidly to a large audience.
Public Awareness and Education:
Helps people understand disaster risks, preparedness steps, and response actions.
Behavioral Change:
Continuous awareness campaigns encourage safer habits and community participation.
Transparency and Accountability:
Keeps government and agencies accountable by reporting on disaster response and relief efforts.
Mobilization of Resources:
Encourages donations, volunteers, and community support for affected people.

Two-way Communication (especially through social media):
Allows people to share information, seek help, and report emergency needs in real time.

Disadvantages of Using Media (4 marks)

Spread of Misinformation or Rumors:
False or exaggerated news can create **panic and confusion** among the public.

Sensationalism:
Some media outlets may dramatize events for higher ratings rather than focusing on facts.

Unequal Access:
Rural or remote areas may have limited access to media or internet connectivity.

Overload of Information:
Too much or conflicting information can make it difficult for people to understand real threats.

Privacy Issues:
Excessive media coverage may invade the privacy of victims or affected families.

(1x10marks =10)